

PANDAS DATA ANALYSIS PROJECT

Overview:

This project performs data analysis using Python and Pandas. The dataset was loaded, analyzed, filtered, and cleaned successfully.

Technologies Used:

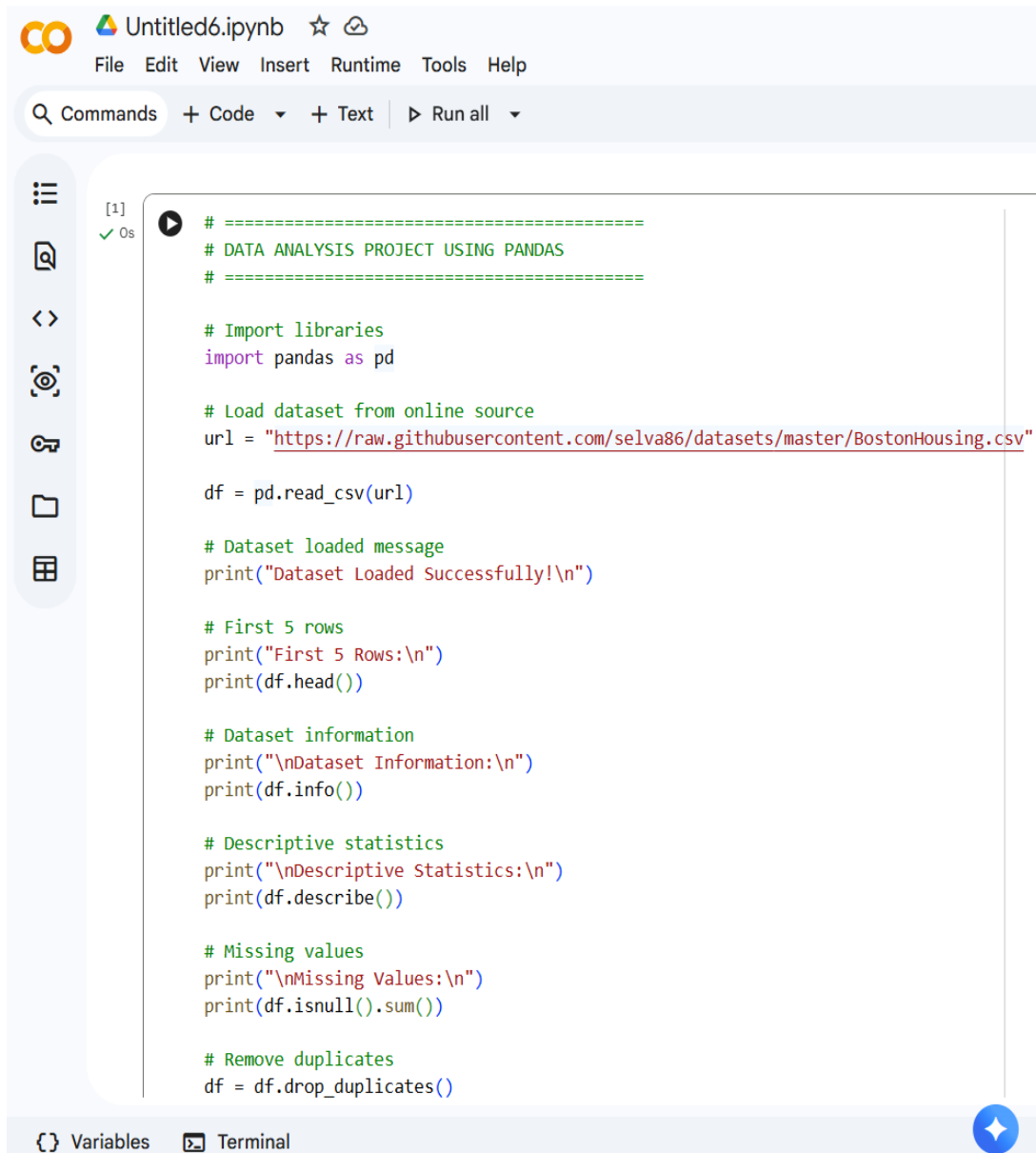
- Python
- Pandas
- Google Colab

Features:

- Data Loading
- Data Cleaning
- Data Filtering
- Data Analysis
- Missing Value Checking
- Duplicate Removal

Screenshots:

- Code Screenshot



The screenshot displays a Jupyter Notebook interface. At the top, the title bar shows 'Untitled6.ipynb' with standard icons for saving, starring, and sharing. Below the title bar is a menu bar with 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help'. A toolbar below the menu bar includes a search icon, 'Commands', and buttons for '+ Code', '+ Text', and 'Run all'. On the left side, a vertical sidebar contains icons for a table of contents, search, expand/collapse, view, key, folder, and a grid. The main area contains a single code cell. The cell's status bar on the left shows '[1]' and a green checkmark with '0s'. The code cell has a play button icon and contains the following Python code:

```
# =====  
# DATA ANALYSIS PROJECT USING PANDAS  
# =====  
  
# Import libraries  
import pandas as pd  
  
# Load dataset from online source  
url = "https://raw.githubusercontent.com/selva86/datasets/master/BostonHousing.csv"  
  
df = pd.read_csv(url)  
  
# Dataset loaded message  
print("Dataset Loaded Successfully!\n")  
  
# First 5 rows  
print("First 5 Rows:\n")  
print(df.head())  
  
# Dataset information  
print("\nDataset Information:\n")  
print(df.info())  
  
# Descriptive statistics  
print("\nDescriptive Statistics:\n")  
print(df.describe())  
  
# Missing values  
print("\nMissing Values:\n")  
print(df.isnull().sum())  
  
# Remove duplicates  
df = df.drop_duplicates()
```

At the bottom of the interface, there is a status bar with a '{ } Variables' button, a 'Terminal' button, and a blue circular button with a white star icon.

Output Screenshot

```
<class 'pandas.core.frame.DataFrame'>
... RangeIndex: 506 entries, 0 to 505
Data columns (total 14 columns):
#   Column      Non-Null Count  Dtype
---  -
0   crim        506 non-null    float64
1   zn          506 non-null    float64
2   indus       506 non-null    float64
3   chas        506 non-null    int64
4   nox         506 non-null    float64
5   rm          506 non-null    float64
6   age         506 non-null    float64
7   dis         506 non-null    float64
8   rad         506 non-null    int64
9   tax         506 non-null    int64
10  ptratio     506 non-null    float64
11  b           506 non-null    float64
12  lstat       506 non-null    float64
13  medv       506 non-null    float64
dtypes: float64(11), int64(3)
memory usage: 55.5 KB
None

Descriptive Statistics:
```

	crim	zn	indus	chas	nox	rm \
count	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000
mean	3.613524	11.363636	11.136779	0.069170	0.554695	6.284634
std	8.601545	23.322453	6.860353	0.253994	0.115878	0.702617
min	0.006320	0.000000	0.460000	0.000000	0.385000	3.561000
25%	0.082045	0.000000	5.190000	0.000000	0.449000	5.885500
50%	0.256510	0.000000	9.690000	0.000000	0.538000	6.208500
75%	3.677083	12.500000	18.100000	0.000000	0.624000	6.623500
max	88.976200	100.000000	27.740000	1.000000	0.871000	8.780000

	age	dis	rad	tax	ptratio	b \
count	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000

Project Files:

- pandas_data_analysis_project.ipynb
- README.md

- Screenshot

Output:

The project successfully displayed:

- First 5 rows
- Dataset information
- Statistical summary
- Missing values
- Filtered data
- Grouped data

Conclusion:

This project demonstrates basic data analysis operations using Pandas in Python.